

Cost Benefit Analysis

- 1 Present Value of a Future Sum
 - reverse interest calculations
 - borrowing against an inheritance
 - considering payout options
- 2 Life Cycle Savings
 - justifying equipment purchases
 - discounted cash flow analysis
- 3 Proposals of Project Topics
 - is a doctoral degree financially worthwhile?
 - use public transport or your own car?

MCS 472 Lecture 18
Industrial Math & Computation
Jan Verschelde, 23 February 2026

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making investment decisions

We need to make an investment decision.

We assume that any investment

- requires sacrificing current resources, and
- has an expected return.

Is the investment worth it?

The main tool to make investment decisions in a cost benefit analysis is a reverse interest calculation.

the present value of a future sum

Consider the following two problems.

- ① In ten years we will inherit \$100,000.
Assuming 5% rate with continuous compounding,
how much can we borrow today?
- ② The lottery (or an insurance settlement) offers
 - ① either to pay half of the prize p now, that is $p/2$; or
 - ② installments of $p/25$ for 25 years.

What should the interest rate be to break even?

To solve these two problems, we need to answer the following.

- ① What is the present value of \$100,000 in ten years?
- ② What is the present value of receiving 25 yearly installments?

a reverse interest calculation

What is the present value of \$3,000 in five years, assuming an annual interest rate of 8%?

$$\$2,042(1.08) = \$2,205$$

$$\$2,205(1.08) = \$2,381$$

$$\$2,381(1.08) = \$2,572$$

$$\$2,572(1.08) = \$2,778$$

$$\$2,778(1.08) = \$3,000$$

Interpretation of the middle number:

$$\begin{array}{ccccc} \$2,572 = & \underbrace{\$2,042(1 + 0.08)^3}_{\text{value of investment}} & = & \underbrace{\$3,000(1 + 0.08)^{-2}}_{\text{value of investment}} \\ & \text{after 3 years} & & \text{2 years before} \end{array}$$

Answer: $\$3,000(1 + 0.08)^{-5} = \$2,042$, rounded to nearest \$.

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borrowing against an inheritance

In ten years we will inherit \$100,000. Assuming 5% rate with continuous compounding, how much can we borrow today?

We start with some notations:

- $t = 10$ is the number of years in the future.
- $P = 100000$ is value of the inheritance.
- $r = 0.05$ is the interest rate.
- L is the amount of the loan.

With continuous interest compounding at rate r , the loan amount L equals P after t years:

$$Le^{rt} = P \quad \text{or} \quad L = Pe^{-rt}.$$

$L = \$100,000e^{-0.5} = \$60,653.07$ is the highest amount of the loan.

the present value of future money

The formula for the present value P_0 of some future money P , paid N years in the future is

$$P_0 = Pe^{-rN},$$

where the interest rate r is called the *discount rate*.

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considering payout options

The lottery (or an insurance settlement) offers

- 1 either to pay half of the prize p now, that is $p/2$; or
- 2 installments of $p/25$ for 25 years.

What should the interest rate be to break even?

What is the present value of the 25 installments?

Let r be the interest rate, using continuous compounding:

$$\underbrace{\frac{p}{25}}_{\text{1st installment}} + \underbrace{\frac{p}{25}e^{-r}}_{\text{in 1 year}} + \underbrace{\frac{p}{25}e^{-2r}}_{\text{in 2 years}} + \cdots + \underbrace{\frac{p}{25}e^{-24r}}_{\text{in 24 years}}.$$

a nonlinear equation

The sum of the 25 installments equals:

$$\frac{p}{25} \left(1 + e^{-r} + e^{-2r} + \dots + e^{-24r} \right) = \frac{p}{25} \left(\frac{1 - e^{-25r}}{1 - e^{-r}} \right).$$

The alternative option is to receive a lump sum of $p/2$.

The break even point is defined by

$$\frac{p}{25} \left(\frac{1 - e^{-25r}}{1 - e^{-r}} \right) = \frac{p}{2}$$

which simplifies into

$$1 - e^{-25r} = 12.5 (1 - e^{-r})$$

so the break even point does not depend on the prize p .

the break even rate

The lottery (or an insurance settlement) offers

- 1 either to pay half of the prize p now, that is $p/2$; or
- 2 installments of $p/25$ for 25 years.

What should the interest rate be to break even?

Answer: 6.74%. (Obtained with NLSolve in Julia.)

The higher the interest rate, the smaller the present value.

using nlsolve

The Julia package `NLsolve` provides a nonlinear solver.

```
using NLsolve
```

```
"""  
    function f!(F, x)  
  
defines F[1] as the input to nlsolve.  
"""  
function f!(F, x)  
    r = x[1]  
    F[1] = 1 - exp(-25*r) - 12.5*(1 - exp(-r))  
end  
  
sol = nlsolve(f!, [0.05])
```

the output of `nlsolve`

```
sol = nlsolve(f!, [0.05])
```

Printing `sol` shows

Results of Nonlinear Solver Algorithm

- * Algorithm: Trust-region with dogleg and autoscaling
- * Starting Point: [0.05]
- * Zero: [0.0673745373743581]
- * Inf-norm of residuals: 0.000000
- * Iterations: 4
- * Convergence: true
 - * $|x - x'| < 0.0e+00$: false
 - * $|f(x)| < 1.0e-08$: true
- * Function Calls (f): 5
- * Jacobian Calls (df/dx): 5

We verify with the evaluation of `sol.zero[1]` using

$f(r) = 1 - \exp(-25*r) - 12.5*(1 - \exp(-r)).$

Is the solution unique?

Exercise 1: Plot the function

$$f(r) = 1 - \exp(-25r) - 12.5(1 - \exp(-r))$$

for a suitable range for r to show that $f(r) = 0$
for more than one value of r .

For what value of the starting point does `nlsolve`
converge to the other solution?

other considerations and perspectives

Exercise 2:

Assuming the installment increases by 3% each year, at which rate does the break even point then occur?

Exercise 3:

Consider the payout plan of the lottery from the state's perspective.

What is the purchase price of q of an annuity that

- 1 pays out $p/25$ for 25 years to the winner,
- 2 the remaining balance on the annuity grows at rate r annually.

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justifying equipment purchases

Consider the purchase of newer, more efficient equipment.

- The purchase requires a present sum of money, a cost.
- The purchase represents a saving, or a benefit.

The cost occurs now, the benefit later.

Compute the justification for the purchase of the new equipment.

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discounted cash flow analysis

The approach to justify equipment purchases is

discounted cash flow analysis.

- 1 For each year of the projected life of the equipment, compute
 - + the projected savings,
 - less costs for that year.

This gives the yearly *net cash flow*.

- 2 Discount the yearly net cash flow back to the present value.
- 3 Compute the *life cycle saving*:
 - + all net cash flow present values over the lifetime,
 - less the first costs.

first costs, return on investment, payback period

Some definitions:

- The *first costs* are the costs for the new equipment.
- The *return on investment* is the discount rate that yields zero life cycle savings.
- The *payback period* is the time before the accumulating present value of the net savings surpasses the initial investment.

the discount rate

We decide to invest in better equipment, which costs q .

The investment

- will save us p
- over n years.

We compute the present value of p at discount rate r :

$$p \left(\frac{1 - e^{-rn}}{1 - e^{-r}} \right)$$

against the cost q of the investment.

The discount rate $r = d - i$, consists of

- 1 d is the interest rate of a safe investment,
- 2 i is the inflation rate.

.

a numerical example

We consider the purchase of new equipment.

- 1 The life span of the equipment is 12 years.
- 2 Each year, the new equipment will save \$1052.
- 3 We use a discount rate $r = d - i = 0.02$.

How much may the equipment cost to justify the purchase?

To answer this question,

- we use a dataframe,
- assuming the equipment costs \$10,000.

See the posted Jupyter notebook.

the solution with a dataframe

	year	savings	costs	values	balance
	Int64	Int64	Int64	Float64	Float64
1	0	0	-10000	0.0	-10000.0
2	1	1052	0	1031.17	-8968.83
3	2	1052	0	1010.75	-7958.08
4	3	1052	0	990.736	-6967.34
5	4	1052	0	971.118	-5996.23
6	5	1052	0	951.889	-5044.34
7	6	1052	0	933.04	-4111.3
8	7	1052	0	914.565	-3196.73
9	8	1052	0	896.455	-2300.28
10	9	1052	0	878.704	-1421.57
11	10	1052	0	861.305	-560.267
12	11	1052	0	844.25	283.982
13	12	1052	0	827.533	1111.51
14	0	12624	-10000	11111.5	0.0

what is the return on investment?

Exercise 4:

Consider the previous numerical example,

- 1 assuming \$7,000 as the purchase cost, and
- 2 inflation rate at $i = 3\%$.

What is the return on investment for this problem?

Exercise 5:

Consider the previous numerical example,
assuming the cost of the equipment is borrowed

- 1 with a 15% down payment,
- 2 at an interest rate of 5%, compounded continuously,
- 3 with a loan over 12 years.

Under these conditions of the loan,
how much may the equipment cost to justify the purchase?

summary and bibliography

Calculating the present value of a future sum,
we provided a quantified justification for investment decisions.

This lecture follows Chapter 7 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

Available online through our UIC library is a cost benefit analysis
description from the management perspective.

- Peter Eichhorn and Ian Towers:
Principles of Management: Efficiency and Effectiveness in the Private and Public Sector. Springer 2018.

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1. Is a doctoral degree financially worthwhile?

Is four years of study beyond the master's to obtain a doctoral degree financially worthwhile?

Consider the following questions:

- Does the added income throughout the career pay back
 - ① the cost of the education, and
 - ② the loss of income during the four years of study?
- Is there is a difference between academia and industry?

2. Quality of a bachelors degree.

What is a good university education?

Consider this question from the point of view of quality, using Taguchi Quality Control.

- 1 Focus on a particular major and define the expectations in financial terms. What is the expected start salary?
- 2 Define the parameters in the quality loss function.
- 3 Gather data and compute the quality loss coefficient.

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3. Use public transport or your own car?

For the daily commute, compare the cost of using public transport versus using your own car.

- In your study consider the normal life span of a car.
- The cost of public transport includes a fixed fare, subject to annual fare hikes.
- The cost of a car includes not only the purchase price, but also taxes, insurance, fuel, repairs, and depreciation cost.

Consider the following questions:

- What is the total saving of using public transport?
- Explain how the annual increase in saving could be used to justify an annual fare hike, that is then also fair...

4. Buy a house or rent?

Suppose you receive a job offer which requires you to move to a different city. Should you buy a house or rent?

- In your study consider a fixed number N of years, which equals the number of years of your job. The N can be used as a parameter.
- Consider the costs and benefits of home ownership.
- Consider the costs and benefits of renting.

Answer the following questions:

- 1 Given the transaction costs of buying a home, what is the minimum value of N to make it worthwhile?
- 2 Home ownership is often cited as a protection against inflation. Consider the scenarios of low and high inflation rate and study the consequences of each in the buying or renting decision.

5. Should we colonize Mars?

What are the benefits of manned missions to Mars?

- 1 Make estimates of the financial investments (first costs) required to put humans on the surface of Mars?
- 2 What are the expected benefits to society?

As a point of departure for your study,
gather historical facts and data of the moon landings.
What were the costs and benefits of the moon missions?